R94B | Small Gain, Limited Recovery

Locked R94B | seed 2 | native/FP64 winners | frozen NEW90k18

01 Completed and locked, not a matched A/B verdict

Actual B controller/worker rc=0; full18 TRAIN/VAL/TEST, LIB and PDFs complete. Selected update 54,720/57,600 (epoch 38), strict minimum full18 VAL macro_vs_sbs_pct; one checkpoint/rule for all datasets. R94A was still running at the request snapshot (~46k/57.6k); not read here. No matched donor-effect claim.

Same frozen NEW90k18 as R89B (main) / R84A (secondary): 81,000/4,500/4,500 per task; seed2, batch1024; preparation hash, native/FP64-ind labels, TRAIN SBS and legal pools verified equal. Eight B tasks keep MTPOMO/MVMOE/reld_mt only; no RELD minima merging. TEST partly informed prior R&D, not a new blind test.

02 Locked full18 metrics

Split	Top1 %	Cost	SBS	Oracle	vsSBS %	vsOrcl %	Regret %
TRAIN	71.9479	11.386290	11.400702	11.345073	-0.2231	0.3617	0.3678
VAL	71.4284	11.416279	11.427978	11.371692	-0.1987	0.3889	0.3991
TEST	71.5778	11.372234	11.385299	11.329932	-0.2112	0.3719	0.3791

ALL = equal-task mean, not pooled raw-cost evidence. SBS is fitted on frozen full TRAIN; Oracle is the legal per-instance minimum. Ratios of means (vsSBS/vsOracle) differ from mean per-instance actual regret.

03 Historical references: positive, but small and uneven

Ref	Split	dTop1 pp	dCost	dRegret pp
R89B	VAL	+0.180247	-0.000015320	-0.001597
R89B	TEST	+0.158025	-0.001236847	-0.010531
R84A	VAL	+0.343210	-0.000332514	-0.004415
R84A	TEST	+0.341975	-0.001609439	-0.015731

R89B TEST: 1,756 corrected - 1,628 harmed = +128 correct; 4,869 predictions changed. Top1 and cost improve on 14/18 tasks; regret improves on 13/18. R84A TEST: +277 net correct. Single-seed historical comparisons do not establish statistical significance or isolate donor pretraining.

04 Fixed36: count true recovery and its price

Fixed R84A VAL zero/near-zero groups (chosen <=0.1%, native support >0); no target reselection. TEST: 117/4,598 true target winners correct (2.5446% recall); 230 selections / 113 false picks (50.8696% precision). VAL: 126/4,625 correct (2.7243%), 112 false picks. Only 12/36 TEST groups have any correct pick; 34/36 have winner support.

Ref	Split	Rescue	Lost	Net	New FP	dTarget R	New-FP R
R89B	VAL	92	18	74	97	-0.001885	0.009442
R89B	TEST	86	14	72	94	-0.006998	0.007458
R84A	VAL	119	4	115	110	-0.004942	0.010947
R84A	TEST	106	2	104	109	-0.010036	0.008097

81/86 TEST rescues vs R89B occur in just four tasks: OVRPB, VRPBL, OVRPBL, OVRPLTW. dTarget R is target-winner regret change, normalized by all task rows and macro-averaged (pp). New-FP R is current regret burden on newly false-selected rows, not a baseline-relative change. Do not add/subtract these subsets; total net dRegret is in section 03.

Decision boundary

Retain B as a small historical gain with limited true recovery. No automatic promotion/variant. Wait for the existing R94A completion before a pretrained-vs-random-donor conclusion. External pretraining adds compute/knowledge; its original instance manifest and overlap remain unknown.

Checkpoint SHA256: 8e797150d22ea138db246536f36a1b7dfa6ef3b516ee065a3303a6b6e579c3ee

Source: completed scalar artifacts only. No GPU, training, live A edits, Git or publication.

R94B | Where the Changes Occur

Locked R94B | seed 2 | native/FP64 winners | frozen NEW90k18

05 All18 TEST changes vs main R89B; secondary R84A shown at right

Task	Top1 %	dTop1 pp	dCost	dRegret pp	R/L/newFP	dTop1 R84
TSP*	47.022	-0.0889	-0.000232	-0.00360	0/2/1	+0.6222
CVRP	48.467	-0.1778	-0.000536	+0.00142	0/0/0	+1.3556
ATSP*	77.178	+0.2667	-1.13e-07	-0.00137	0/0/0	+0.4667
OVRP	40.467	-0.0222	-0.000762	-0.01260	2/3/12	-0.2444
VRPB	81.067	+0.3778	-0.000850	-0.01030	0/0/0	+0.4222
VRPL	37.667	-0.6444	+0.000678	+0.00568	0/0/0	+0.5111
VRPTW	81.356	+0.1778	-0.004746	-0.02799	1/0/2	+0.2000
OVRPTW	84.089	+0.1556	+0.001230	+0.01340	1/0/1	+0.0889
OVRPB	88.333	+0.1111	-0.001141	-0.02489	18/4/17	+0.2222
OVRPL	41.800	+0.1778	+0.000061	+0.00285	0/0/0	-0.0444
VRPBL	80.422	+0.9556	-0.003998	-0.04505	22/0/22	+0.8667
VRPBTW	79.111	+0.3111	-0.004779	-0.03061	0/0/0	+0.2444
VRPLTW	79.489	+0.0667	-0.000774	-0.00428	1/1/5	+0.0889
OVRPBL	89.511	+0.2444	-0.001500	-0.02808	19/3/20	+0.2000
OVRPBTW	85.089	+0.1556	-0.001128	-0.01031	0/0/0	+0.1556
OVRPLTW	83.267	+0.2667	+0.001051	+0.01509	22/1/14	+0.3333
VRPBLTW	80.267	+0.4000	-0.003487	-0.01785	0/0/0	+0.5333
OVRPBLTW	83.800	+0.1111	-0.001350	-0.01105	0/0/0	+0.1333

* TSP/ATSP are raw-only. VRPL loses Top1/cost; OVRPTW, OVRPL and OVRPLTW gain Top1 but worsen cost/regret. Full per-task metrics, SBS/Oracle, both references and recovery losses: round_analysis_by_problem.csv.

06 Coverage and full-TRAIN fitting, not an old tiny-cohort story

Covered16 TEST vs R89B: dTop1 +0.166667 pp, dCost -0.001376926, dRegret -0.011536 pp; 86 rescued / 12 lost. Raw2: dTop1 +0.088889 pp, dRegret -0.002487 pp; 0 rescued / 2 lost. Covered16 VAL cost worsens +0.000082329 despite the tiny full18 VAL cost gain. Raw-only changes are not direct donor evidence.

Selected full TRAIN vs VAL Top1: 71.947874% vs 71.428395%, gap 0.519479 pp; VAL-TRAIN regret +0.031272 pp. TRAIN target recall is only 3.2781% vs VAL 2.7243%. No large tiny-subset memorization claim or regularization-cause inference is supported.

07 LIB / reused NSS: mixed, not uniform transfer superiority

Dataset	N	B Top1 %	NSS Top1 %	B cost	NSS cost
TSP TEST	4500	47.0222	45.8889	7.970665	7.970401
CVRP TEST	4500	48.4667	45.9778	18.009808	18.004802
TSPLIB	49	24.4898	20.4082	8.212695	8.217907
CVRPLIB	100	64.0000	70.0000	67.140632	67.101112

NSS = reused R79 historical weights/predictions, NEW4500 TEST plus original LIB; no NEW90k retrain. Same B checkpoint/pools throughout. TSPLIB vs R89B: Top1 -2.0408 pp, cost +0.010187; CVRPLIB: Top1 -1 pp, cost -0.097105. Vs R84A, TSPLIB Top1 +4.0816 pp but cost +0.003230; CVRPLIB Top1 +1 pp and cost -0.109740. NSS has lower cost on both synthetic TEST tasks and CVRPLIB.

08 Timing call-path error; saved predictive metrics not implicated

r94_experiment.py:327-334 reads FrozenFeatureCache.batch, then engine.inputs_only(batch). r94_features.py:414-435 injects nco_nodes/nco_node_mask; pairwise_objective.py:10-14 removes labels but retains both. r94_model.py:191-212 executes the donor only when those fields are absent. Thus the nominal donor+selector path uses injected cached features on covered tasks; TSP/ATSP never use a donor. All five inspected source hashes match the recorded protocol; a CPU-only dictionary-helper check confirms identical raw/cached keys and objects.

Reported differences (0.000046 to 0.000737 ms/instance) are NOT measured donor overhead. Online donor latency is unestablished; includes_native_encoder / donor_plus_selector_measured metadata is unsupported. Only GPU-resident cached-selector batch timing remains: 128 rows, one warmup, three repetitions, excluding I/O/cache preparation. No end-to-end claim, rerun, runtime fix or result change here.